Mexico's yield curve: does it follow a pattern, can it be forecast, and does the pattern help forecast it?

Question 1. Is there a pattern in how the nine maturities move? Yes. The curve moves in three typical ways. In the **level**, **all rates rise or fall together**. In the **slope**, short and long rates move differently, so the gap between them widens or narrows. In the **curvature**, the middle maturities move differently from the ends. Over three months or more, the three movements explain more than 90% of the changes. **The slope follows Banxico's policy rate, and since 2012 the level has also moved with the US.**

Question 2. Can the curve be forecast one and four quarters ahead? Short-term rates (CETES), yes. One quarter ahead, their slope anticipates Banxico's rate moves and cuts the error by 12% to 35% compared with assuming the rate stays unchanged. Four quarters ahead the curve alone is not reliable, but **Banxico's survey of analysts shows that moving CETES by the change analysts expect in Banxico's rate cuts the error** by about 22%, in 2013–2019 and in 2020–2026. **Long-term rates (BONOS M), no, not even with the US bond, survey expectations, sovereign risk or the exchange rate**; they also depend on the risk premium, which is not directly observed.

Question 3. Does the pattern improve the forecasts?

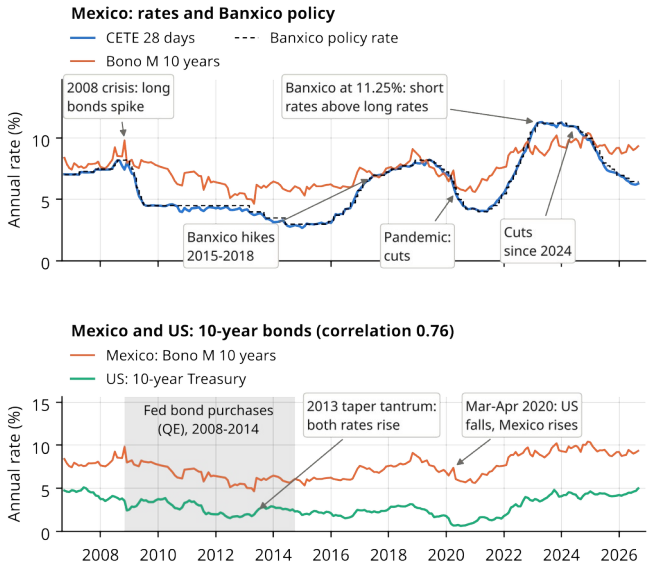
Yes, when it is used as a signal of what **the market expects Banxico to do with its policy rate**, which is what the gap between the one-year and the 28-day rate captures.

No, when each movement is assumed to revert to its historical average, because that average has shifted with each Banxico cycle. Neither external data nor special treatment of crises changes this result; Banxico's survey helps one year ahead, but it is information from outside the curve.

1. The data and their context

Twenty years of data show that **the 28-day CETES rate practically matches Banxico's policy rate** (its target rate), while the 10-year bond moves more independently, spiking in the 2008 crisis and falling below short rates in 2023–2024, when Banxico raised its rate to 11.25% (Figure 1, top panel). In addition, **the Mexican and US 10-year bonds move together, especially since 2012** (Figure 1, bottom panel). These two features, short maturities following Banxico's policy rate and long ones following that rate and also the US bond, recur throughout the analysis.

Figure 1. Twenty years of the curve: short rates follow Banxico and long rates also respond to the US



Source: monthly averages of primary auctions, Banco de México (Oct-2006 to Sep-2026), checked against Banxico's weekly auction records (section 1 of the notebook). Policy rate (target rate): Banxico, SIE (SF61745); before 2008, weighted funding rate (SF43773). Treasury bond: FRED (DGS10), monthly average. QE (*quantitative easing*): large-scale Fed bond purchases to lower US long rates. *Taper tantrum*: in May–June 2013 the Fed announced it would scale back those purchases, and long rates rose in the US and in emerging markets, including Mexico.

The curve is built from the monthly auction rates of nine federal government instruments, CETES from 28 to 364 days and BONOS M from 3 to 30 years, used as observed.[1] **The file matches Banxico's official figures in 99% of its 1,936**

[1]The 10-, 20- and 30-year bonds are not auctioned every month (about 30% of months are missing); for estimation those gaps were filled with the trend between neighboring months, and forecasts were scored only against months with an actual auction. CETES are sold at a discount and BONOS M pay coupons every six months, so their rates are not fully comparable; the technical ("zero-coupon") curve would require prices and coupons for each issue, which are not in the data (Diebold and Li, 2006).

monthly values; three discrepancies were corrected, including a small upward bias in the 364-day CETE, without changing any conclusion (section 1 of the notebook).

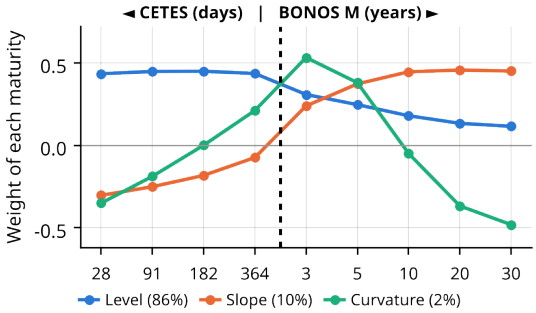
2. Question 1: the pattern of the curve

Analyzing how the rates of the nine maturities change (rather than their level, because all of them share a long-term trend that would inflate the result[2]) reveals the three typical movements that the literature has documented in many bond markets.[3] How much they explain depends on the period over which changes are measured. **In one-month changes, the three movements (level, slope and curvature) together explain 78%**; **that they do not explain more appears to be because each bond is auctioned in different weeks**, so the monthly average mixes different moments and adds noise.

In three-month changes they explain 91%, and in one-year changes, 99%. This is because **over longer periods, brief movements and auction noise cancel out**, leaving mostly what persists, such as Banxico's cycles (periods in which it raises or lowers its policy rate continuously, like the 2015–2018 hikes or the 2019–2021 cuts), which move all maturities in an orderly way. Splitting the sample into two ten-year halves, the three movements have almost the same shape in both, but their weight changes: in 2006–2015 the level explained 63% and the slope 30%; in 2016–2026, the level 91% and the slope 7%, because **Banxico's recent cycles moved the whole curve in the same direction** (Table 1).

How much each movement weighs. Of all annual changes in the nine rates, **86% correspond to movements in which the whole curve rises or falls together**, that is, in which maturities behave alike; 10% to movements in which the gap between short and long rates widens or narrows; and 2% to movements in which the middle maturities (3- and 5-year bonds) move differently from the ends (Figure 2). The remaining 2% are small movements with no clear pattern.

Figure 2. Weight of each maturity in the curve's three movements (one-year changes)



Each line is a movement and shows how much each maturity takes part; in parentheses, the share of annual changes it explains. The dashed line separates CETES (28 to 364 days) from BONOS M (3 to 30 years).

Why each movement has the shape in Figure 2. The level weighs more on CETES because **short rates follow Banxico's policy rate almost one for one**, while **long rates average the expected long-run return and move less**. The slope is negative for CETES and positive for BONOS M, and almost constant from 10 to 30 years because those maturities respond together to common factors, such as US rates and the risk premium. The curvature peaks at the 3- and 5-year bonds, which reflect what is expected from Banxico's policy rate over the next few years, so they rise more than the ends when the market anticipates a hiking cycle that will not last, as in 2021–2022.

Episodes and periods

Outliers. Extreme movements concentrate in three episodes: the financial crisis (October 2008 and February 2009), the *taper tantrum* (June 2013) and the pandemic (March 2020). In those months the level of the curve changed by more than 1.1 points, when the typical change is 0.2 points and 95% of months stay below 0.7. The literature identifies the same episodes as periods of stress for emerging-market debt.[4] Extreme movements were not removed from the analysis; section 4 shows that keeping them improves the forecasts. The curvature's largest jump came instead at the start of the 2021–2022 hikes, when the market priced a steep cycle it did not expect to last, as explained above (3-year bond +3.8 points in a year; 28-day CETE +2.3; 30-year bond +0.9).

Inverted curve. For 43 months the curve's fitted slope was negative, that is, short rates were above long rates, mostly in 2023–2024, when Banxico raised its rate to 11.25%; the slope hit its minimum in April 2023 (-3.4 points). Under the expectations hypothesis, this signals that the market expects short rates to fall,[5] and Banxico did start cutting in March

[2] Rate levels share long trends that make the first component explain 94% "by construction". Standard practice is therefore to analyze changes (Litterman and Scheinkman, 1991).
[3] Litterman and Scheinkman (1991) identified the three movements in the US; they have since been documented in many bond markets (Diebold and Rudebusch, 2013).
[4] In 2009 and 2020 the level's volatility was three to five times that of the calmest years. See Aguilar, Diego-Fernández, Elizondo and Roldán-Peña (Banxico, WP 2020-18) on Mexico's term premium, Eichengreen and Gupta (2015) on 2013 and Hördahl and Shim (2020, BIS Bulletin 18) on March 2020.
[5] Under the expectations hypothesis, a long rate is roughly the average of the short rates the market expects, plus a premium. Campbell and Shiller (1991) show that the **gap between long and short rates anticipates the direction of short rates: when it is negative, they tend to fall.**

2024; section 3 builds on this mechanism.

Economic interpretation. The slope of the curve follows Banxico's policy rate in every period, with a correlation between -0.82 and -0.94 (Table 1). The relationship is negative because, when Banxico raises its rate, short rates rise more than long rates and the gap between them narrows. This relationship is stronger than the one Banxico found for 2001–2007 (-0.65).[6] The level, by contrast, began moving with US rates from 2012, in line with recent Banxico studies finding that **global factors weigh increasingly on Mexico's long rates**.[7]

Table 1. Periods of highest (2022–2026) and lowest (2012–2016) similarity across maturities: the level weighs more when Banxico changes its rate by large amounts

Period	What was happening	Share of annual changes in which the whole curve moves together	Did the slope move with Banxico's rate?	Did Mexico's level rise and fall with the 10-year US bond?
2007–2011	Crisis; Banxico cuts its rate from 8.25% to 4.5%	77%	Yes (-0.92)	No (-0.10)
2012–2016	Banxico rate between 3% and 4.5% until 2015; *taper tantrum*; Fed hikes	51% (lowest similarity)	Yes (-0.82)	Yes (0.71)
2017–2021	Banxico raises its rate to 8.25% and cuts it to 4% in the pandemic	91%	Yes (-0.91)	Yes (0.66)
2022–2026	Banxico raises its rate to 11.25% and cuts it from 2024	93% (highest similarity)	Yes (-0.94)	Partly (0.46)

Note: the numbers in parentheses are correlations (near 1 or -1, the series move together; near 0, unrelated). Fourth column, the curve's fitted slope against Banxico's rate, negative because the slope narrows when Banxico hikes. Fifth column, annual changes in Mexico's level against those of the 10-year US bond.

3. Question 2: forecasting the curve

Evaluation design

"One quarter ahead" means the rate three months later; "four quarters", twelve months later. To reproduce real conditions, in each month from December 2012 every model was estimated only with the information available up to that month and a forecast was issued, which gave 163 forecasts at three months and 154 at twelve.

Benchmark: assuming the rate stays unchanged. Every forecast was compared with the simplest assumption, that the rate in three or twelve months will equal today's (in the literature, the "random walk"). **For interest rates this benchmark is usually hard to beat**; many studies find that term-structure models, and more recently machine-learning methods, improve on it only slightly or only at some maturities.[8]

Models of increasing complexity were tested, each to check one idea. An autoregressive model (AR), which forecasts each rate from its own recent values, checks whether a rate's history is enough. A second model adds a single variable to the no-change assumption, the CETES slope, to check whether what the market expects from Banxico anticipates rates. A vector autoregression (VAR), which forecasts each rate from the recent values of all nine, checks whether some maturities anticipate others. Three models that summarize the curve in its three movements answer question 3.[9]

Results

The result splits according to the segment of the curve (Table 2 and Figure 3). **For CETES at three months, adding the slope reduces the error** by 25% on average, between 12% and 35% depending on maturity, more the shorter it is; the nine-rate model does about the same (28%), because the gap between the one-year and the short rates is already among its nine rates.

For BONOS M no model beats the no-change assumption, and those that summarize the curve in three movements err up to 42% more. **At twelve months the CETES error falls 22% in 2020–2026**, but in 2013–2019 it rises 6%, so the result is not reliable. Twenty years of data also contain few independent one-year periods (section 4 shows that Banxico's survey solves this for CETES).

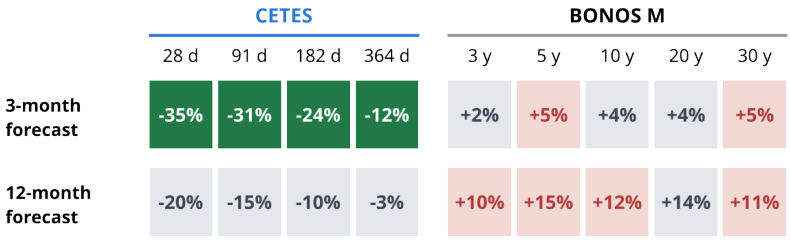
The economic explanation is the expectations hypothesis, according to which the gap between the one-year rate and the 28-day rate reflects what the market expects Banxico to do with its rate in the coming months.[10] That is why the

[6] Cortés, Ramos-Francia and Torres (Banxico, WP 2008-07) and Cortés and Ramos-Francia (Banxico, WP 2008-09), with zero-coupon curves for 2001 to 2008; Banco de México, Quarterly Report Apr–Jun 2022, Box 6, on the pass-through of the policy rate to 1-month to 10-year rates.
[7] Aguilar et al. (*Latin American Economic Review*, 2024); Banco de México, Quarterly Report Jul–Sep 2025, Box 5: a global factor explains more than 75% of Mexican long rates' variation in 2020–2025.
[8] A model that does not beat the random walk adds no information about the future. It is the standard benchmark (Duffee, 2002; Diebold and Li, 2006). Kang (2026), with US bonds and machine learning, finds that the gains concentrate at short maturities and decline as the horizon lengthens; for Mexico, Elizondo (Banxico, WP 2013-03) found it was the best 12-month forecast.
[9] The AR model forecasts each rate from its current value and assumes it reverts to its average (one lag). All models forecast directly at 3 or 12 months, which makes them more robust to regime changes (Marcellino, Stock and Watson, 2006); the Clark and West (2007) test confirms the contribution of the slope.
[10] See note 5; Fama and Bliss (1987) document the same predictive power. For Mexico, Ibarra (Banxico, WP 2021-07) finds related evidence for economic activity.

slope anticipates moves in that rate, while assuming the rate stays unchanged only recognizes them once they have happened (Figure 4).

The error reduction concentrates in quarters when Banxico moved its rate by more than half a point, such as the 2020 cuts and the 2021–2022 hikes, where the error falls 35%; when Banxico held its rate, the model erred more than the benchmark. Long bonds depend on factors outside today's curve, such as US rates (section 4) and the risk premium, which was not included because it is not observed.[11]

Figure 3. Slope model versus assuming the rate stays unchanged: the error falls only for CETES at 3 months



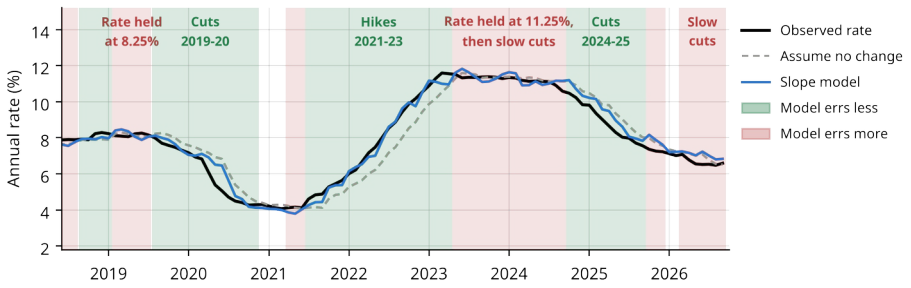
Change in the error of the slope model relative to assuming the rate stays unchanged. Each column is the rate being forecast; each row, the horizon. Green: errs less; red: errs more; grey: not significant (Diebold-Mariano, 95%). 163 forecasts at 3 months and 154 at 12, from Dec-2012.

Table 2. Error of each model compared with the no-change assumption

Model	3-month forecast (one quarter)			12-month forecast (four quarters)		
	CETES	BONOS M	Curve	CETES	BONOS M	Curve
Each rate's history (AR)	1.04	1.02	1.03	1.05	1.05	1.05
No change + CETES slope	0.75	1.04	0.91	0.88	1.12	0.95
All nine rates together (VAR)	0.72	1.02	0.88	0.89	1.11	0.95
Three movements + VAR (Nelson-Siegel)	0.90	0.99	0.94	0.96	1.06	0.99
Three movements + AR (Diebold-Li)	1.21	1.03	1.12	1.14	1.10	1.13
Three movements + AR (principal components)	1.13	1.42	1.28	1.04	1.12	1.06
Error of the no-change assumption (percentage points)	0.60	0.58	0.59	2.09	1.20	1.69

How to read: "Curve" covers all nine maturities. Each number is the model's typical error divided by the error of assuming the rate stays unchanged (root mean squared error, RMSE). Below 1: the model errs less (0.75 = 25% less error). Above 1: it errs more. Green or red: significant at 95%, that is, not attributable to chance (Diebold-Mariano test).[12] No color: small or unstable difference.

Figure 4. 91-day CETE: the slope model errs less while Banxico is moving its rate and more while it pauses



Each point on the forecast lines is what was expected for that month three months earlier. Shading compares the absolute errors of the two forecasts (three-month average): green, the slope model errs less; red, it errs more. Labels show what Banxico was doing.

4. Question 3: does the pattern improve the forecasts?

It depends on how it is used. Each model that uses the pattern was compared, over the same months, with an equivalent one that forecasts each rate from its own past values only (Table 3). The pattern helps when it is used as a signal of what the market expects Banxico to do with its rate, because **the CETES slope summarizes what the market expects from Banxico's policy rate** and, with a single variable, matches a model with all nine rates and 90 parameters.

The pattern hurts when it is used only to describe the curve and pull each movement back to its past average. The Diebold-Li model assumes each of the three movements reverts to its full-sample average. But **in Mexico that average has shifted with each Banxico monetary policy cycle**: the policy rate went from 3% in 2015 to 8.25% in 2018, to 4% in 2021 and to 11.25% in 2023. It therefore forecasts reversions that do not happen.[13]

[11] It requires a term-structure model, which would add its own error; observed proxies (US bond, EMBIG spread) were tested instead.
[12] Diebold and Mariano (1995) with the Harvey, Leybourne and Newbold (1997) correction; errors adjusted for overlapping horizons.
[13] Duffee (2002) reports the same problem in mean-reverting models, and Pesaran and Timmermann (2007) in models estimated on data that include regime changes.

Table 3. Change in error of models that use the pattern, compared with forecasting each rate from its own past values

Model that uses the pattern	3-month forecast	12-month forecast
CETES slope	-12%	-10%
Nelson-Siegel + VAR (the three movements influence each other)	-8%	-6%
Diebold-Li (each movement reverts to its average)	+9%	+7%
CETES slope, compared instead with all nine rates together (VAR)	+3%	-1%

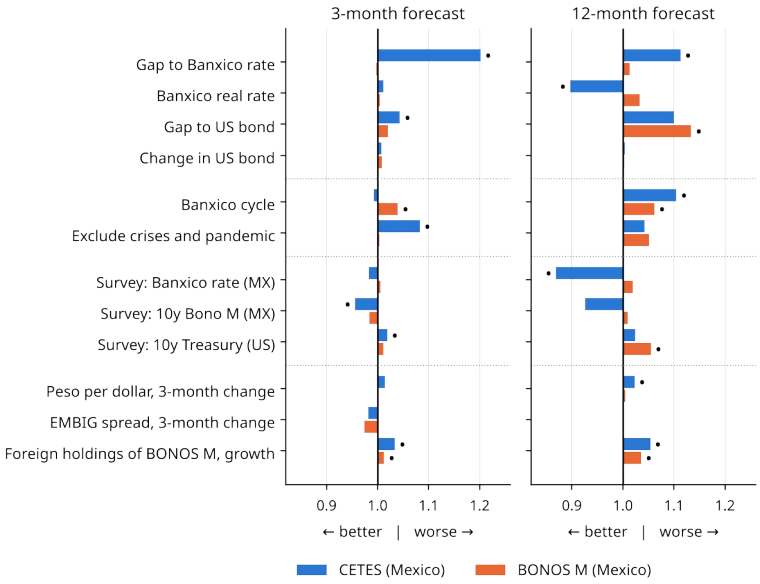
Negative: the model with the pattern errs less. Green or red: significant at 95% (Diebold-Mariano); no color: small or unstable.

Does information from outside the curve help?

Five kinds of variants were tested (Figure 5): Banxico policy data, the US bond, crisis treatment, survey expectations and market risk.[14] **No variant reliably beats the slope model. Only Banxico's real rate (policy rate minus inflation) improves CETES at twelve months** (10% less error), but the improvement is fragile, because it comes **only from the 2020–2026 cycle** and disappears in 2013–2019. Excluding crises from estimation worsens the forecast, because those months show how the curve reacts when Banxico changes its rate by large amounts.[15]

Banxico's survey of analysts is the only addition that makes the one-year CETES forecast reliable. Adding the change analysts expect in Banxico's rate cuts the slope model's CETES error at twelve months by 13% (Figure 5); even with no estimation, moving each CETES rate by that change errs about 22% less than assuming no change, in 2013–2019 (-18%) and 2020–2026 (-24%). At three months it adds nothing to the slope, because both capture the same near-term expectations. For BONOS M, the survey's expected 10-year yield cuts the one-year error by only 5%, not significant, and the US survey worsens it, for the same reason as the US bond. **The exchange rate does not improve the forecasts either**, nor do sovereign risk (EMBIG spread) or foreign holdings of BONOS M; none of them beats the no-change assumption for BONOS M.[16]

Figure 5. Of the variants shown, only Banxico's survey of analysts improves on the slope model, and only for CETES at one year



Error of each variant ÷ error of the slope model, same forecast dates; CETES in blue, BONOS M in orange. Left of 1, the variant improves; the dot marks significance at 95%. Survey rows add the expected change in the rate shown (median; Banxico from 2010 and 2013, Philadelphia Fed SPF); the last three rows add the market variables of note 16.

5. Conclusion

The Mexican curve moves mostly as a block (86% of annual changes). **Its slope, although it weighs less, reflects what the market expects from Banxico and anticipates short rates one quarter ahead**, especially when Banxico changes its rate by more than half a point in a quarter. One year ahead, CETES are best forecast with what analysts expect from Banxico; for long bonds, the best bet is still no change. What it adds to understanding the market is that **CETES reveal what the market expects from Banxico, while long bonds respond to global factors outside the local curve.**

[14] Surveys: Banxico's Survey of Private Sector Economists and the Philadelphia Fed's Survey of Professional Forecasters. Each data point was used once public (policy rate at month end; inflation with a one-month lag; Banxico's survey from the following month; the US survey from mid-quarter). Excluded episodes: Oct-2008 to Jun-2009, May–Sep 2013 and Mar–Aug 2020. Altavilla, Giacomini and Ragusa (2017) find that surveys improve forecasts mostly at the short end, as here.

[15] Pesaran and Timmermann (2007): under regime changes, keeping earlier data tends to improve forecasts.

[16] EMBIG Mexico spread, peso per dollar and foreign holdings of BONOS M (sources in the notebook), each added alone. The peso change moves the error by less than 3%; the EMBIG level and foreign holdings worsen it. The three-month EMBIG change (-2% at three months) and a factor of five rates (Cochrane and Piazzesi, 2005; -5% to -9% for BONOS M) beat the slope model but not no change.